

An RTOS-Based AI-Enabled Fitness Wearable with Edge-Cloud Hybrid Intelligence

ABSTRACT

Commercial fitness trackers rely on proprietary firmware, while academic projects typically use Arduino super-loops on breakout boards, lacking real-time determinism and intelligent interaction. This project addresses these gaps by designing CoachOS — a wrist-worn device on a student-fabricated four-layer SMD PCB with an ESP32-C3, MAX30102 PPG sensor for heart rate and SpO₂, LIS3DH accelerometer for step counting, 1.3-inch round touchscreen, and MEMS microphone with audio amplifier. A milled sensor tab ensures optical coupling, while conformal coating and a silicone gasket protect exposed electronics.

The software runs FreeRTOS with eight prioritized tasks: sensor sampling, heart rate extraction, step detection, data fusion, LVGL touchscreen UI, and tickless idle power management. A novel “Hey Coach” voice assistant uses on-device wake-word detection and local command recognition, while complex queries stream via WiFi to a Node.js backend with OpenAI Whisper, GPT-4o-mini with live sensor context, and text-to-speech. This edge-cloud hybrid balances real-time response with AI capability on a constrained device. Targets include ± 3 BPM heart rate accuracy, $\pm 5\%$ step count, six-hour workout battery life, and sub-five-second cloud latency. Deliverables comprise the functional prototype, ESP-IDF firmware, cloud backend, and technical documentation.

REFERENCES

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Project Team

1. Shahazad Abdulla
2. Tharun Oommen Jacob
3. Sidharth Sajith

Er. Jittu George Eapen
Project Guide